MIL-G-21196B
20 February 1968
SUPERSEDING
MIL-G-21196A
26 April 1961
(See 6.3)

MILITARY SPECIFICATION

GAGES, CYLINDER

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers two types of cylinder gages.

1.2 Classification. Cylinder gages shall be of the following types, as specified (see 6.2):

Type I - Dial mounted on block with locking device and slide base.
Type II - Dial mounted on slide base with setting gage.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

MILITARY
MIL-H-15424 - Hand Tools; Packaging of.
MIL-I-18422 - Indicators, Dial and Accessories.

STANDARD

MILITARY
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL BUREAU OF STANDARDS PUBLICATION
Handbook H28 - Screw Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Material. Material used in the construction of cylinder gages and accessories shall be as specified hereinafter.

3.2 Design. The cylinder gage shall be composed essentially of the components of parts specified herein for each type.

3.3 Dial indicator. The dial indicator shall conform to the requirements of MIL-I-18422. Type I cylinder gages shall have type III, style B indicators. Type II cylinder gages shall have type III, style A indicators. Accessories shall be manufacturer's standard.

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3.4 Alignment slide base. The alignment slide base shall be of a forged steel, hardened, ground and so designed that when the cylinder gage is set in the cylinder wall, it shall be capable of sliding over the entire area to be measured without perceptible shake or play, making the gage self-centering.

3.5 Accessories. Cylinder gages shall be provided with extension rods and contact points. (See 3.10.4 and 3.11.2.)

3.6 Screw threads. Screw threads on cylinder gages and accessories shall conform to Handbook H28.

3.7 Heat treatment. All parts of the cylinder gage such as slides, gage blocks and contacts which may affect the accuracy through wear shall have a minimum hardness of 55 on the Rockwell "C" scale (except indicator parts).

3.8 Finish. All finished exterior surfaces of the cylinder gages, except the gaging surfaces, shall have a surface roughness not to exceed 63 microinches arithmetical average (A.A). The gaging surfaces shall have a surface roughness not to exceed 8 microinches (A.A).

3.9 Identification and marking. Cylinder gages and accessories that require it shall be marked in a permanent and legible manner with the manufacturer's name or trademark of such known character that the source of manufacture may be readily determined and with manufacturer's catalog or other identifying number.

3.10 Type I, dial mounted on block, with locking device. Type I cylinder gages shall consist essentially of a dial indicator, mounting block, alignment slide base with fixed contact point, locking device, toggle joint handle and extension rods which form a point of contact. Type I cylinder gage shall be similar to figure 1 and shall have a range of either 2-1/10 to 6 inches, 2-1/2 to 6 inches, or 2-1/2 to 9 inches as specified (see 6.2).

3.10.1 Mounting block. The mounting block shall be the manufacturer's standard. It shall be of uniform quality, free from blow holes, porosity, hard spots, shrinkage, or any other imperfections that may impair its serviceability.

3.10.2 Locking device. The locking device shall consist of a steel, knurled head, lock screw, which protrudes above the dial and screws into a threaded hole in the body of the mounting block. It shall lock the adjustable contact point, which is synchronized with the indicating needle on the face of the dial, by a double spring action in the assembly.

3.10.3 Toggle joint handle. The handle shall be of steel, hollow, cylindrical in shape, and knurled on the outer surface. One end shall be provided with a knurled cap screw and the opposite end shall be provided with a threaded section that assembles into the toggle joint which is fitted in the section of the cylinder block. The handle shall have a diameter of approximately 5/16 inch and a length of 8-1/2 inches.

3.10.4 Extension rods. The extension rods shall be hardened and ground steel and shall slide into an opening in the lower section of the cylinder block. The rods shall be securely held in the gage spindle by a method that provides an easy means for removing and inserting the rods. The rods shall be approximately 5/32 inch in diameter and shall be of such lengths to provide for measuring all 0.001 inch increments within the range of the gage. When maximum cylinder bore diameter is specified (see 6.2), extra extension rods shall be furnished with the 2-1/2 to 9 inch range gage for measuring bores over 9 inches in diameter.

3.11 Type II, dial mounted on slide base, without locking device. Type II cylinder gages shall consist essentially of a dial indicator, alignment slide base, tool setting handle, contact points and extension rods. The gage shall have a range of 2 to 6 inches.

3.11.1 Tool setting handle. Tool setting handle shall be made of steel rods and assembled in two sections. One section of the handle shall be threaded on one end and the opposite end fitted into an opening provided in the top section of the alignment slide base. The other section of the handle shall have a fixed spacer, a fixed anvil, a removable anvil and a set of seven finished and lapped spacers. The spacers shall be assembled on the handle to provide the approximate size of bore to be measured by the use of proper setting of the spacers. When specified (see 6.2), a plain handle in lieu of a tool setting handle shall be furnished.

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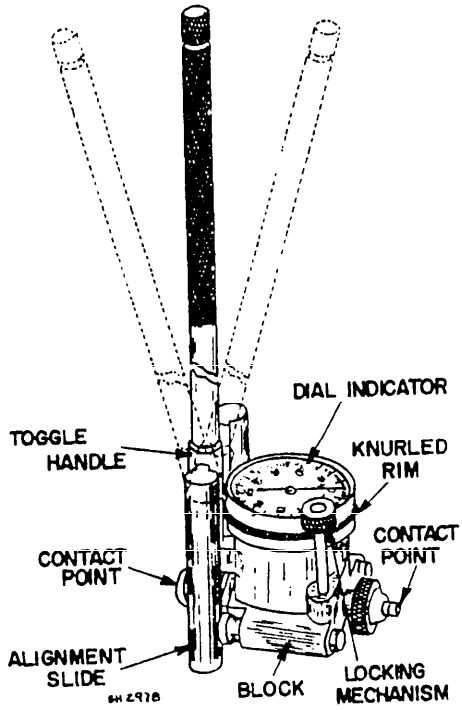


Figure 1. Type I cylinder gage with dial mounted on block, with locking device.

3.11.2 Extension rods and contact points. Cylinder gages shall be provided with four extension rods and two contact points of steel, hardened and ground. Extension rods shall be threaded on one end and an internal thread on the opposite end. Contact points shall be threaded on one end only. The threads provide for assembly into the rack of the indicator. The outer surface of the extension rods and contact points shall have a knurled grip.

3.11.3 Case. The case for type II cylinder gage gages, when specified (see 6.2) shall be in accordance with the latest manufacturing practice. The case shall be of metal or wood and of sufficient strength and durability to withstand normal usage. The case shall be so designed as to house completely, and hold securely, the complete gage and accessories.

3.11.4 Type II gages and accessories shall be similar to figures 2, 3 and 4.

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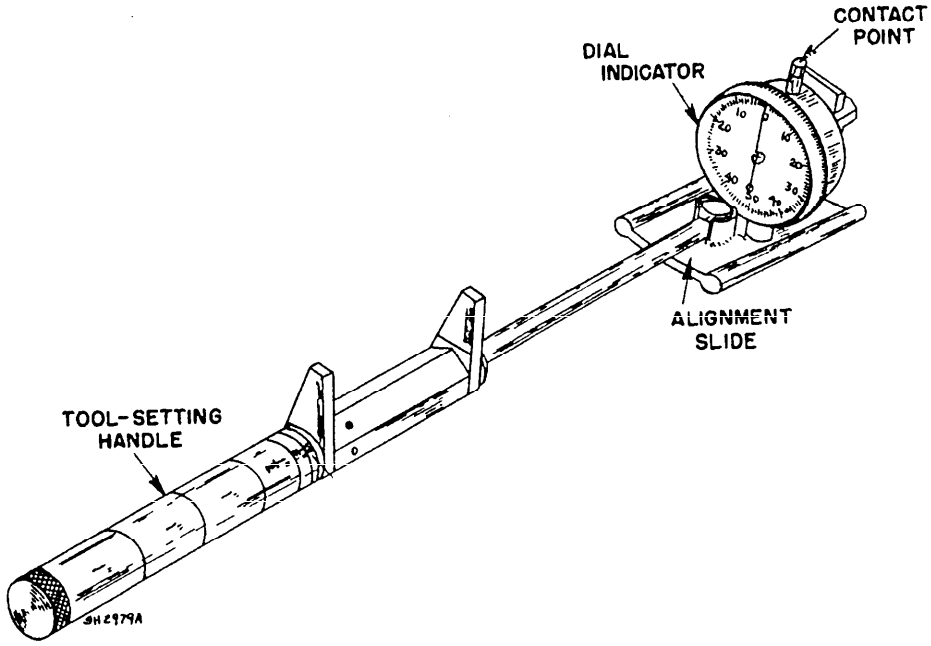


Figure 2. Type II cylinder gage with dial mounted on slide base, without locking device.

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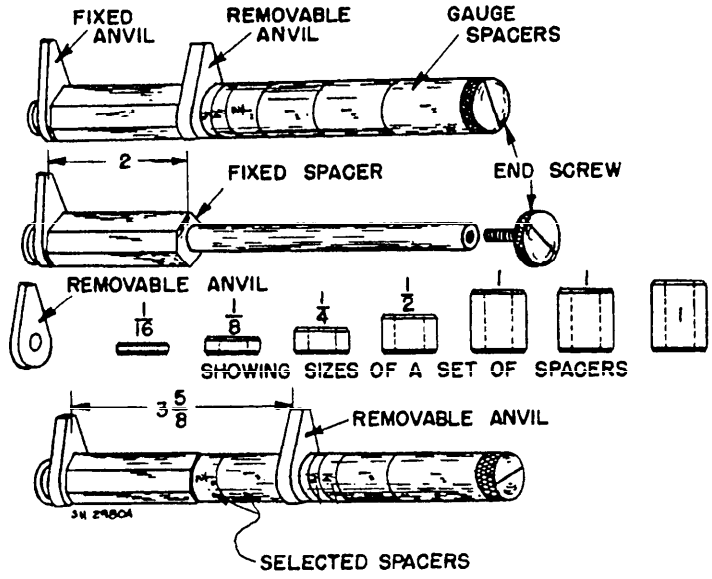


Figure 3. Tool setting handle for type II cylinder gage.

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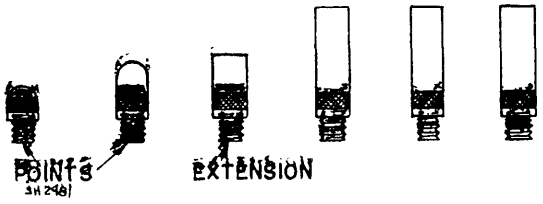


Figure 4. Extension rods and contact points for type II cylinder gages.

3.12 Workmanship. Workmanship shall be of the quality prevailing among manufacturers normally producing cylinder gages of the type specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Sampling for quality conformance.

4.2.1 Lot. All cylinder gages of the same type having the same capacity range, and offered for delivery at one time, shall be considered a lot for purposes of quality conformance inspection.

4.2.2 Sample. A random sample shall be selected from each lot in conformance with MIL-STD-105 at inspection level III.

4.2.3 Quality conformance inspection. Quality conformance inspection shall be performed on each sample gage selected in accordance with 4.2.2. Quality conformance inspection shall consist of the following:

(a) Examination (see 4.3).
(b) Tests (see 4.4).
(c) Inspection of preservation and packaging (see 4.5).

4.3 Examination. Each of the sample gages selected in accordance with 4.2.2 shall be examined to determine conformance with this specification. Any gages in the sample containing one or more visual or dimensional defects shall be rejected, and if the number of defective gages in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected. The acceptable quality level shall be 1.5 percent defective.

4.4 Test procedures. Each of the sample cylinder gages selected in accordance with 4.2.2 shall be tested to determine conformance with the requirements of this specification. Any gage in the sample containing one or more defects shall be rejected, and if the number of defective gages exceeds the acceptance number for that sample, the lot represented by that sample shall be rejected. The acceptable quality level shall be 4.0 percent defective.

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4.4.1 Hardness. Each slide, gage block, contact and anvil shall be subjected to a hardness test to determine conformance with the requirements of this specification.

4.4.2 Surface roughness. Surface roughness shall be determined at representative spots by commercially recognized instruments for surface roughness measurement.

4.4.3 Dial indicators shall be tested in accordance with test requirements of MIL-I-18422.

4.5 Inspection of preservation, packaging, packing, and marking for shipment and storage. Preservation, packaging, packing, and marking for shipment and storage shall be examined to determine conformance with the documents referenced in Section 5. The acceptable quality level shall be 4.0 percent defective.

5. PREPARATION FOR DELIVERY

(Preparation for delivery requirements specified herein apply only for direct Government procurements. Preparation for delivery requirements of referenced documents listed in Section 2 do not apply unless specifically stated in the contract or order. Preparation for delivery requirements for products procured by contractors shall be specified in the individual orders.

5.1 Preservation, packaging, packing, and marking. Unless otherwise specified (see 6.2), preservation, packaging, packing, and marking shall be in accordance with MIL-H-15424. Levels of preservation, packaging and packing shall be as specified (see 6.2).

6. NOTES

6.1 Intended use. The gages covered by this specification are intended for use in the measurement of cylindrical holes for straightness, out of roundness, and taper diameters of engine cylinders.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type of cylinder gage required (see 1.2).
(c) Range required (see 3.10, 3.10.4 and 3.11).
(d) When plain handle shall be furnished in lieu of tool setting handle for type II (see 3.11.1).
(e) When case is required (see 3.11.3).
(f) Specify levels of preservation, packaging, and packing and marking requirements (see 5.1).

6.3 CHANGES FROM PREVIOUS ISSUE. THE OUTSIDE MARGINS OF THIS DOCUMENT HAVE BEEN MARKED "#" TO INDICATE WHERE CHANGES (DELETIONS, ADDITIONS, ETC.) FROM THE PREVIOUS ISSUE HAVE BEEN MADE. THIS HAS BEEN DONE AS A CONVENIENCE ONLY AND THE GOVERNMENT ASSUMES NO LIABILITY WHATSOEVER FOR ANY INACCURACIES IN THESE NOTATIONS. BIDDERS AND CONTRACTORS ARE CAUTIONED TO EVALUATE THE REQUIREMENTS OF THIS DOCUMENT BASED ON THE ENTIRE CONTENT AS WRITTEN IRRESPECTIVE OF THE MARGINAL NOTATIONS AND RELATIONSHIP TO THE LAST PREVIOUS ISSUE.

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SPECIFICATION ANALYSIS SHEET

Form Approved
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INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

1 APR 63 FORM 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

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